

医药翻译PRD.pdf - 纯译文

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Medical Translation PRD

Prototype reference document is continuously being updated at <http://10.3.75.127:3000/>

I. Project Background

1.1 Business Background

The pharmaceutical industry frequently deals with multilingual document processing in daily R&D, regulatory submissions, academic exchanges, and international collaborations. Typical scenarios include:

- Bilingual translation of clinical trial reports, investigator brochures, and informed consent forms;
- Creation of multilingual versions for medical device technical documents, instructions, and labels;
- Rapid translation and side-by-side review of pharmaceutical academic papers and regulatory submissions (e.g., FDA, EMA, NMPA).

Currently, users mainly rely on general-purpose translation tools or manual translation. General tools lack specialized pharmaceutical terminology support, leading to inaccurate and inconsistent translations of key terms. Manual translation is costly and time-consuming, failing to meet the demand for frequent, small-batch, real-time translations. Moreover, existing tools poorly preserve formatting in Word and PDF documents—requiring extensive manual reformatting after translation, severely impacting efficiency.

1.2 User Pain Points

Pain Point	Description
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Inaccurate Terminology	General translation engines fail to recognize pharmaceutical terms like "randomized controlled trial", "primary endpoint", and "adverse event", resulting in unusable translations.
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Formatting Easily Broken	After translating PDF or Word files, tables, headers/footers, and figure layouts become disordered, requiring significant manual correction.
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Lack of Parallel Versions	Only translated text is provided without a quick way to compare it against the original; bilingual versions must be manually assembled.
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No Historical Traceability	Each translation requires re-uploading and re-setting language preferences, making it impossible to retrieve previous translation results.
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Multilingual configuration is cumbersome—source and target languages must be manually selected each time, resulting in low efficiency.

Data security concerns—pharmaceutical documents contain sensitive information, and using public translation tools poses a risk of data leakage.

Inconvenience in handling multiple files—after uploading several documents at once, there is no unified management or side-by-side preview capability.

1.3 Technical Feasibility

- A hybrid approach combining machine translation and a pharmaceutical terminology database has been maturely applied in the industry, ensuring high accuracy of specialized terms.
- Document format preservation technologies (such as Apache POI, Aspose, and PDF parsing engines) can achieve stable layout output for Word/PDF files.
- Single Sign-On (SSO) combined with user-space isolation solutions meet enterprise-level security requirements.

2. Product Value

2.1 User Value

Value Point	Description
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Accurate and consistent terminology	Built-in pharmaceutical terminology database ensures uniformity of terms throughout the
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document and compliance with industry standards—ready for use in official documents.

Zero formatting adjustments | Translated Word/PDF files retain original layout (allowing minor length adjustments while preserving direction and structure)—no manual corrections required.

Significant efficiency improvement | Instant response for text translation; batch processing for files; one-click reuse of historical records—average translation time reduced by over 80%.

Convenient comparison and review | Automatically generates side-by-side preview for translated documents, offering both pure translation and bilingual comparison versions for download to meet different scenarios.

Full traceability | All translation tasks are automatically saved, supporting search, filtering, and quick re-execution—preventing redundant work.

Data security and isolation | SSO + user data isolation ensures confidentiality and compliance of pharmaceutical documents.

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Intuitive operation — Task-oriented interface, clear at a glance, no training required.

2.2 Business Value

Value Point | Description

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Reduce translation costs | Machine translation replaces most manual translation, reducing outsourcing expenses; estimated annual cost reduction of 60% – 70%.

Accelerate business processes | Real-time translation capability shortens processing time for drug registration documents and clinical trial documents from days to minutes.

Enhance international competitiveness | Rapid generation of high-quality multilingual documents supports enterprise expansion into overseas markets.

Knowledge asset accumulation | Historical records form an enterprise-specific pharmaceutical translation corpus, which can improve terminology databases and support future AI training.

3. Requirements

3.1 User Authentication and Data Isolation

- As a user, I want the system to prioritize Single Sign-On (e.g., Feishu) to use unified corporate identity authentication without remembering multiple account credentials.

Acceptance Criteria:

1. The system login page should prominently display a "Feishu Login" button; clicking it redirects users to the Feishu authorization page.

2. If Feishu SSO is unavailable due to configuration issues, the system must provide a backup username/password login form.

3. After successful login via either method, the user enters the main system interface with an active session that persists until manually logged out or timed out.

- As a user, I want the system to implement data isolation so that each user can only view and operate their own translation records, ensuring data security and privacy.

Acceptance Criteria:

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1. After User A logs in, the left-side history list displays only User A's translation tasks and shows no records from other users.

2. User A cannot access or download User B's translation results by directly entering a URL or any other method.

3. The system shall perform backend validation based on the logged-in user 's identity for all data operations to ensure data isolation security.

3.2 General Language Configuration

- As a user, I want the system to support at least bidirectional translation between Chinese and English, and as many common languages as possible (e.g., Japanese, Korean, German, French), to handle multilingual translation scenarios.

Acceptance Criteria:

1. In both text translation and file translation modes, two dropdown menus for source language and target language should be provided above the input area, allowing users to freely select any supported language.
2. The source language dropdown must include options “ Auto Detect, ” “ Chinese, ” and “ English, ” with “ Auto Detect ” as the default. When the user inputs Chinese text, the system should automatically recognize it as “ Chinese ” ; when inputting English text, it should automatically recognize it as “ English, ” and display the recognized result in the dropdown. The same applies to file uploads.
3. If the user manually changes the source language to a specific language (e.g., “ English ”), the system shall use the user-selected language for translation and no longer override it automatically.
4. The target language dropdown does not support auto-detection; the user must manually select the target language. The enumeration must include at least “ Chinese ” and “ English. ”

3.3 Medical Industry General Terminology Library Mounting

- As a user, I want the system to call the medical industry general terminology library during translation to ensure accurate and consistent professional term translations.

Acceptance Criteria:

1. Upon submission of a translation request (text or file), the system backend must invoke the medical industry general terminology library for term matching and replacement.
2. For medical terms included in the terminology library (e.g., "randomized controlled trial"), the translation must uniformly use the standardized translation from the library (e.g., "randomized controlled trial")—no non-library direct translations are allowed.

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3. If a term is not found in the terminology database, the system shall use a general machine translation engine to process it normally, without interrupting the translation due to missing terms.

3.4 General Translation Requirements

As a user, I want to see a translation status prompt (e.g., "Loading", "Success", or "Failure") after clicking "Send", so I can understand the translation progress. When failure occurs, retry functionality should be supported.

Acceptance Criteria:

1. After the user clicks "Send", the button should immediately become grayed out, navigate to the result page, and display either "Translating..." or a loading animation to prevent duplicate clicks.
2. Upon successful translation completion, the original text should be displayed on the left side of the page and the translated text on the right side, with a "Translation Successful" message shown briefly before automatically disappearing.
3. If translation fails due to network or server issues, an explicit "Translation Failed" message should be displayed, along with a "Retry" button in the translation area. When the user clicks "Retry", the system should resend the translation request.
4. After a successful retry, the status should update to "Success"; if the retry still fails, the "Retry" button should reappear, with no limit on the number of retries.

3.3.1 Text Translation

As a user, I want to input medical text in text translation mode so that I can quickly complete short texts or segments' translation tasks.

Acceptance Criteria:

1. When the user is in text translation mode, a multi-line text input box should appear on the right side, allowing the user to enter medical text up to a maximum of 5000 characters.
2. After the user inputs the text and clicks the "Send" button, the system should immediately begin processing the translation.

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3. During the translation process, the "Send" button should immediately be disabled and display "Translating..." or a loading animation until the translation is complete (whether successful or failed), at which point the button becomes usable again.
- As a user, I want the translation result to appear directly on the right side of the page (not in a tooltip bubble) so that I can clearly view both the original text and the translated text, and copy the translation content.

Acceptance Criteria:

1. After text translation is completed, no tooltip-style bubble should appear on the right side of the page; instead, clear cards or areas should separately display "Original Text" and "Translated Text."
2. The original and translated texts must be clearly labeled (e.g., with headings “ Original Text ” and “ Translated Text ”) to facilitate comparison.
3. Next to the translated text area, there should be a "Copy" button or icon.
4. When the user clicks the copy button, the translated text should be copied to the system clipboard, followed by a brief prompt saying "Copied."
5. The copied content must be complete, in plain text format, without extra spaces or line breaks.

As a user, I want text translation to support only single-turn queries without maintaining multi-turn conversation context, so that I can focus on individual translation tasks without interference.

Acceptance Criteria:

1. After completing one text translation, if the user enters new text and clicks send again, the system should treat this as a completely new and independent task, with no contextual connection to previous original or translated text (e.g., it should not automatically refer to the subject of the previous sentence).
2. Each translation result should be displayed independently on the right side, without stacking or showing historical translations (historical records are saved separately in the left-side list).

3.3.2 File Translation

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3.4 Results Display and Download

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3.5 Historical Record Management

3.6 Interface and Experience

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3.3 Non-functional Objectives

Dimension | Objective

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Performance | Text translation response ≤ 2 seconds; File translation response ≤ 30 seconds (for documents up to 10 pages)

Availability | System annual availability $\geq 99.5\%$

Security | Data encrypted at rest and in transit using HTTPS; user sessions isolated

Compatibility | Supports latest two versions of Chrome and Edge; resolution 1280×720 or higher

Maintainability | Terminology database can be updated independently without releasing a new version